

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5501

5 Credits: 4

Program Core

Source-grounded

Linear Integrated Circuits

□ To provide an insight on the fundamental concepts of operational amplifiers and their

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 5501 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5501.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Electronics and Computer Engineering
Course code	5501
Course title	Linear Integrated Circuits
Semester	5 Credits: 4
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester:60
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD**How to use this lesson****First pass**

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide an insight on the fundamental concepts of operational amplifiers and their applications.
- To give a functional description of various linear integrated circuits and discuss on various circuits which act as the bridge between analog and digital domains.

Course outcomes

CO	Official outcome	Study level
CO1	12 Understanding electrical parameters of operational amplifier.	See official syllabus
CO2	Develop circuits using operational amplifier. 20 Applying	See official syllabus
CO3	Implement circuits using timer IC and PLL 12 Applying Explain the operation of various fixed and	See official syllabus
CO4	14 Understanding variable voltage regulators, ADC and DAC	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 3
CO3	CO3 3

Row	Extracted official mapping
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	operational amplifier.	4	As specified
Module 2	Develop circuits using operational amplifier.	5	As specified
Module 3	Implement circuits using timer IC and PLL	4	As specified
Module 4	ADC and DAC	3	As specified

MODULE 1**operational amplifier.**

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: operational amplifier.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 4 Understanding amplifier Explain the block diagram of operational

4 Understanding amplifier Explain the block diagram of operational is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding amplifier Explain the block diagram of operational using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding amplifier explain the block diagram of operational should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Understanding amplifier Explain the block diagram of operational means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 4 Understanding amplifier Explain the block diagram of operational definition/meaning. steps, application. Technical terms English- Malayalam.

▼ 2 Understanding amplifier

2 Understanding amplifier is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding amplifier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding amplifier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding amplifier means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- 2 Understanding amplifier ഏകദേശം
 ഏകദേശം definition/meaning ഏകദേശം. ഏകദേശം ഏകദേശം ഏകദേശം, steps,
 application ഏകദേശം ഏകദേശം ഏകദേശം. Technical terms English- ഏകദേശം
 ഏകദേശം.

▼ Explain operational amplifier parameters

Explain operational amplifier parameters is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain operational amplifier parameters using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain operational amplifier parameters should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain operational amplifier parameters means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Explain operational amplifier parameters
 പരമ്പരീകൃതമായും പരിശോധിക്കുക definition/meaning പരമ്പരീകൃതമായും. പരമ്പരീകൃതമായും
 പരമ്പരീകൃതമായും, steps, application പരമ്പരീകൃതമായും പരമ്പരീകൃതമായും. Technical terms
 English- പരമ്പരീകൃതമായും പരമ്പരീകൃതമായും.

▼ Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower

Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower is an official topic in Module 1 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational

Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the basic op-amp feedback circuits 3 understanding fundamentals of operational amplifier differential amplifier: bjt differential amplifier - types - large and small signal operation - input resistance - voltage gain - CMRR - non-ideal characteristics operational amplifier : op-amp symbol - package types - block diagram - dc and ac op-amp parameters - equivalent circuit - voltage transfer curve - open loop op-amp configurations - op-amp with negative feedback - properties of practical op-amp - inverting amplifier - non-inverting amplifier - voltage follower should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation - Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation - Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□. □□□□□ □□□□□ □□□□□□□□, steps, application □□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Develop circuits using operational amplifier.

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop circuits using operational amplifier.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

▼ Develop various amplifiers using op-amp

Develop various amplifiers using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop various amplifiers using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop various amplifiers using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop various amplifiers using op-amp means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓപ്റ്റോണൽ ഡെവലപ്പ് വാരിയസ് അംപ്ലിഫയർസ് ഉപയോഗിച്ച് ഓപ്റ്റോണൽ ഡെവലപ്പ് വാരിയസ് അംപ്ലിഫയർസ്. ഓപ്റ്റോണൽ ഡെവലപ്പ് വാരിയസ് അംപ്ലിഫയർസ്, steps, application ഓപ്റ്റോണൽ ഡെവലപ്പ് വാരിയസ് അംപ്ലിഫയർസ്. Technical terms English-ൽ ഓപ്റ്റോണൽ ഡെവലപ്പ് വാരിയസ് അംപ്ലിഫയർസ്.

▼ Developwave-shaping circuits using op-amp

Developwave-shaping circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Developwave-shaping circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, developwave-shaping circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Developwave-shaping circuits using op-amp means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Developwave-shaping circuits using op-amp □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

▼ Develop oscillator circuits using op-amp

Develop oscillator circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop oscillator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop oscillator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop oscillator circuits using op-amp means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-ഓസിലേറ്റർ സർക്യൂട്ട് ഉപയോഗിച്ച് ഓസിലേറ്റർ സർക്യൂട്ട് വികസിപ്പിക്കുക. ഓസിലേറ്റർ സർക്യൂട്ട് definition/meaning ഓസിലേറ്റർ സർക്യൂട്ട്. ഓസിലേറ്റർ സർക്യൂട്ട് ഓസിലേറ്റർ, steps, application ഓസിലേറ്റർ സർക്യൂട്ട് ഓസിലേറ്റർ. Technical terms English-ഓസിലേറ്റർ സർക്യൂട്ട് ഓസിലേറ്റർ സർക്യൂട്ട്.

▼ Illustrate comparator circuits using op-amp

Illustrate comparator circuits using op-amp is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate comparator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate comparator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate comparator circuits using op-amp means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Illustrate comparator circuits using op-amp □□□□□□□□□□ □□□□ definition/meaning □□□□□□□□□□. □□□□□□ □□□□□□, steps, application □□□□□ □□□□□□□□ □□□□□. Technical terms English-□ □□□□ □□□□□□□□□□.

▼ **Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters.**

Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. is an official topic in Module 2 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference

amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, developactive filters using op-amp 4 applying applications of operational amplifier summing amplifier - difference amplifier - scaling and averaging amplifiers - adder - subtractor - instrumentation amplifier - voltage to current converter, current to voltage converter - integrator - differentiator - precision rectifiers - peak detector - clipper and clamper - log and antilog amplifier - phase shift and wien bridge oscillators - astable and mono stable multi vibrators - triangular and saw tooth wave generators- comparators - zero crossing detector - schmitt trigger - first order low-pass, high-pass and band-pass butterworth filters. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave

Aspect	What to prepare
	generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ Developactive filters using op-amp 4 Applying Applications of operational amplifier Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier - Voltage to current converter, Current to voltage converter - Integrator - Differentiator - Precision rectifiers - Peak detector - Clipper and Clamper - Log and antilog amplifier - Phase shift and Wien bridge oscillators - Astable and mono stable multi vibrators - Triangular and saw tooth wave generators- Comparators - Zero crossing detector - Schmitt trigger - First order low-pass, high-pass and band-pass Butterworth filters. □□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Implement circuits using timer IC and PLL

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Implement circuits using timer IC and PLL
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ 3 Understanding timer

3 Understanding timer is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding timer using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding timer should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding timer means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding timer ഏകദേശം നിർവ്വചനം definition/meaning നിർവ്വചനം. ഏകദേശം നിർവ്വചനം, steps, application നിർവ്വചനം നിർവ്വചനം നിർവ്വചനം. Technical terms English- നിർവ്വചനം നിർവ്വചനം.

▼ Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565

Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop circuits using 555 timer ic 3 applying explain the functional block diagram of 565 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565 definition/meaning, steps, application Technical terms English-

▼ 3 Understanding PLL

3 Understanding PLL is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding PLL using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding pll should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding PLL means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 3 Understanding PLL ഫീഡ്ബാക്ക് സർക്യൂട്ട് ഫീഡ്ബാക്ക് definition/meaning ഫീഡ്ബാക്ക് സർക്യൂട്ട്. ഫീഡ്ബാക്ക് സർക്യൂട്ട് ഫീഡ്ബാക്ക്, steps, application ഫീഡ്ബാക്ക് സർക്യൂട്ട് ഫീഡ്ബാക്ക്. Technical terms English- ഫീഡ്ബാക്ക് ഫീഡ്ബാക്ക്.

▼ **Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators,**

Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, is an official topic in Module 3 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing

Explain the operation of various fixed and variable voltage regulators, using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate applications of PLL 3 understanding timer & PLL 555 timer IC - functional block diagram - features - a stable and mono stable multi vibrator using 555 - phase locked loop - general block diagram of PLL - operation - lock range, capture range and pull-in time - functional block diagram of PLL IC 565 - applications of PLL for AM & FM detection and frequency multiplication, frequency division, frequency synthesizing explain the operation of various fixed and variable voltage regulators, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC - Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോന്നും Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators, ഓരോന്നും definition/meaning ഓരോന്നും. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നും. Technical terms English-ഓരോന്നും ഓരോന്നും.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

ADC and DAC

This module is presented from the official syllabus scope for Linear Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: ADC and DAC
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to

Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain fixed and variable voltage regulators 4 understanding compare and contrast various analog to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to definition/meaning, steps, application Technical terms English-

▼ 5 Understanding digital converters Compare and contrast various digital to

5 Understanding digital converters Compare and contrast various digital to is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding digital converters Compare and contrast various digital to using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding digital converters compare and contrast various digital to should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 5 Understanding digital converters Compare and contrast various digital to means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 Understanding digital converters
Compare and contrast various digital to Δ converters
definition/meaning Δ converter. Δ converter Δ converter, steps,
application Δ converter Δ converter. Technical terms English- Δ Δ
 Δ .

▼ **5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/117107094/> 2 <https://nptel.ac.in/courses/117101106/> 3 <https://nptel.ac.in/courses/108/108/108108111/> 4 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 5 <https://freevideolectures.com/course/2915/linear-integrated-circuits>**

5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e,

2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/117107094/> 2 <https://nptel.ac.in/courses/117101106/> 3 <https://nptel.ac.in/courses/108/108/108108111/> 4 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 5 <https://freevideolectures.com/course/2915/linear-integrated-circuits> is an official topic in Module 4 of Linear Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-back protection- optocoupler - principle of operation. Data Converters: D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers

and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008
 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1
 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010
 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3
 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford
 University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link
 1 <https://nptel.ac.in/courses/117107094/> 2
<https://nptel.ac.in/courses/117101106/> 3
<https://nptel.ac.in/courses/108/108/108108111/> 4
<https://inst.eecs.berkeley.edu/~ee140/fa19/> 5
<https://freevideolectures.com/course/2915/linear-integrated-circuits> using
 standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 5 understanding analog converters ic voltage regulators and data converters monolithic voltage regulators: fixed voltage regulators, 78xx and 79xx series - adjustable voltage regulators ic lm723 - low voltage and high voltage configurations, current boosting, current limiting, short circuit and fold-backprotection- optocoupler - principle of operation. data converters:d/a converter - specifications - weighted resistor type, r-2r ladder type. a/d converters: specifications - classification - flash type - counter ramp type - successive approximation type - dual slope type - sample-and-hold circuits. text / reference: t/r book title/author roy d. c. and s. b. jain, linear integrated circuits, new age international, t1 3/e, 2010 sergio franco, design with operational amplifiers and analog integrated t2 circuits, 3rd edition, tata mcgraw-hill, 2008 gayakwad r. a., op-amps and linear integrated circuits, prentice hall, r1 4/e, 2010 r2 botkar k. r., integrated circuits, 10/e, khanna publishers, 2010 salivahanans. ,v. s. k. bhaaskaran, linear integrated circuits, tata r3 mcgraw hill, 2008 david a. bell, operational amplifiers & linear ics, oxford university r4 press, 2nd edition,2010 online resources: sl.no website link 1

<https://nptel.ac.in/courses/117107094/> 2 <https://nptel.ac.in/courses/117101106/>
 3 <https://nptel.ac.in/courses/108/108/108108111/> 4
<https://inst.eecs.berkeley.edu/~ee140/fa19/> 5
<https://freevideolectures.com/course/2915/linear-integrated-circuits> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 https://nptel.ac.in/courses/117107094/ 2 https://nptel.ac.in/courses/117101106/ 3 https://nptel.ac.in/courses/108/108/108108111/ 4 https://inst.eecs.berkeley.edu/~ee140/fa19/ 5 https://freevideolectures.com/course/2915/linear-integrated-circuits means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-back protection- optocoupler - principle of operation. Data Converters: D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 Salivahanan S., V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition, 2010 Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/117107094/> 2 <https://nptel.ac.in/courses/117101106/> 3 <https://nptel.ac.in/courses/108/108/108108111/> 4 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 5 <https://freevideolectures.com/course/2915/linear-integrated-circuits> ഫലപ്രദമായ പദപരിചയം definition/meaning പദപരിചയം. പദപരിചയം പദപരിചയം പദപരിചയം, steps, application പദപരിചയം പദപരിചയം പദപരിചയം. Technical terms English-ൽ പദപരിചയം പദപരിചയം.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL**Module-wise questions and answers**

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► **Define or explain 4 Understanding amplifier Explain the block diagram of operational. (3 marks)**

► **Define or explain 2 Understanding amplifier. (3 marks)**

► **Define or explain Explain operational amplifier parameters. (3 marks)**

► **Define or explain Illustrate the basic op-amp feedback circuits 3 Understanding Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types - Large and small signal operation - Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier : Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower. (3 marks)**

Module 2

► **Define or explain Develop various amplifiers using op-amp. (3 marks)**

► **Define or explain Develop wave-shaping circuits using op-amp. (3 marks)**

► **Define or explain Develop oscillator circuits using op-amp. (3 marks)**

► **Define or explain Illustrate comparator circuits using op-amp. (3 marks)**

Module 3

► **Define or explain 3 Understanding timer. (3 marks)**

► **Define or explain Develop circuits using 555 timer IC 3 Applying Explain the functional block diagram of 565. (3 marks)**

► **Define or explain 3 Understanding PLL. (3 marks)**

► **Define or explain Illustrate applications of PLL 3 Understanding Timer & PLL 555 Timer IC- Functional block diagram - features - a stable and mono stable multi vibrator using 555 - Phase Locked Loop - General block diagram of PLL - Operation - Lock range, capture range and pull-in time - Functional block diagram of PLL IC 565 - Applications of PLL for AM & FM detection and Frequency multiplication, Frequency division, Frequency synthesizing Explain the operation of various fixed and variable voltage regulators,. (3 marks)**

Module 4

► **Define or explain Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to. (3 marks)**

► **Define or explain 5 Understanding digital converters Compare and contrast various digital to. (3 marks)**

► **Define or explain 5 Understanding analog converters IC Voltage regulators and Data Converters Monolithic Voltage Regulators: Fixed**

voltage regulators, 78XX and 79XX series - Adjustable voltage regulators IC LM723 - Low voltage and high voltage configurations, Current boosting, Current limiting, Short circuit and Fold-backprotection- optocoupler - principle of operation. Data Converters:D/A converter - Specifications - Weighted resistor type, R-2R Ladder type. A/D Converters: Specifications - Classification - Flash type - Counter ramp type - Successive approximation type - Dual slope type - Sample-and-hold circuits. Text / Reference: T/R Book Title/Author Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T1 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T2 circuits, 3rd Edition, Tata McGraw-Hill, 2008 Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R1 4/e, 2010 R2 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R3 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R4 Press, 2nd edition,2010 Online Resources: Sl.No Website Link 1 <https://nptel.ac.in/courses/117107094/> 2 <https://nptel.ac.in/courses/117101106/> 3 <https://nptel.ac.in/courses/108/108/108108111/> 4 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 5 <https://freevideolectures.com/course/2915/linear-integrated-circuits.> (3 marks)

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **4 Understanding amplifier**
Explain the block diagram of operational.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Develop various amplifiers using op-amp.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **3 Understanding timer.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **Explain fixed and variable voltage regulators 4 Understanding Compare and contrast various analog to.**

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://nptel.ac.in/courses/117107094/>
<https://nptel.ac.in/courses/117101106/>
- <https://nptel.ac.in/courses/108/108/108108111/>
<https://inst.eecs.berkeley.edu/~ee140/fa19/>
- <https://freevideolectures.com/course/2915/linear-integrated-circuits>

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